Section: CS May 30, 2026

Multi-Signature Authentication Web Proxy

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Problem Statement

Background and statement of the theoretical or practical problem, issue or opportunity motivating the research.

Enterprise grade cryptocurrency wallets have long implemented multi-signature ("multi-sig") authentication, which requires two or more parties to sign an outbound transaction from a secure wallet before it is sent to the receiving party. This security mechanism ensures that before a transaction is sent, it is agreed upon by the group, rather than any one individual. But what if you want to require multiple parties to approve of a critical setting change or release a package? No existing general-purpose web authentication system (think Duo or Authelia) requires multiple distinct people to cooperate before granting access.

My practicum project proposes a general-purpose multi-signature authentication web proxy that allows access to the requested resource only when multiple parties provide their secret keys.

Supply chain attacks have been wreaking havoc on developers in recent months. Current solutions focus on locking down individual developer accounts, but I believe a better solution is requiring multiple human verifiers to approve a project release before it goes public. My project will provide a solution that, while generalizable, will be directly applicable to this use case. This project will demonstrate this gap and then advocate for platform-native functionality to be added to the repositories.

Solution Statement

This research will (the overall aim of the research) in order to (how the research will address the problem) with specific work product/deliverables described.

This research will create a general-purpose multi-signature authentication web proxy that, as an example of a real-world use case, will prevent any one individual from publishing a package to PyPI and instead require multiple reviewers to approve of the package's release. I will develop an application that will be hosted on a web server. When a developer attempts to publish a package to PyPI, the web proxy will:

1. Intercept the request while saving it for publishing 2. Notify other maintainers that a package is attempting to be published 3. Gather enough approvals from the maintainers until a quorum is reached 4. Then forward the publish request on to the PyPI repository

Completed Tasks (Last 2 Weeks)

- I brainstormed project ideas and created the initial Project Proposal.

- I continued to define the projects scope, motivation, use cases, and deliverables and incorporated them into the first progress report.
- I conducted research into real-world use cases for this project to better define the project's motivation.
- I defined a use case, publishing packages through PyPi, with a real world data flow. This use case research with AI-assisted summary can be found here.
- I utilized AI to create a broad literature review on topics discussed in my project proposal for future reference. This broad literature review research AI-assisted report can be found here.
- I began targeted research into multi-signature authentication cryptography primitives.

Tasks for the Next Project Report

What will you do over the next two weeks for your project?

1. Decide on a multi-sig authentication scheme.
    - Secrets range from username/password combinations to tokens, how will these signatures be stored, and how will they be able to be used in a multi-signature scheme?
    - Must be based on research peer-reviewed white papers
2. Create in-depth, researched application specifications
    - Flowchart from client machine → web proxy → notification to approvers → approvals → access allowed/denied to secure endpoint
    - Authentication system specification using multi-signature authentication scheme
    - Web proxy application specification
    - Approver notification system specification
    - Approval system specification

Questions I Have or Issues I'm Running Into

Are there unanticipated obstacles, is something taking longer than you thought it might, or is there an area where the Instructional Team might be able to provide some direction or clarification?

When researching use cases, I found it difficult to conceptualize how I'm going to integrate multi-signature authentication into already existing platforms. For example, for the PyPI use case, the existing authentication system in PyPI is that you have an owner and multiple maintainers. The owner and the maintainers can all push packages to the public repository any time they wish. There is no way to restrict this functionality. This becomes a problem for my solution because I need to have fine-grained controls over who can push to the public repository. The solution I found is a bit of a hack. For the Web Proxy to be the only one with access to publishing rights on the repository, there will need to be a service account specific to the user's project that is the owner of the repository and there needs to be no maintainers. Then my web proxy will be the only one with the publishing token. Ideally, the solution is for every repository to have this multi-signature function. Therefore, my research and deliverables serve more as a proof-of-concept and is instead built to show that this functionality is possible and should be a first-class authentication mechanism in package repositories. By making my project general use, this advocacy for multi-sig authentication functionality could be expanded to other use cases which I have not yet identified.

Methodology Paragraph Summary

Description of the methodology you are proposing to get from the starting point of your project to the final deliverable. Projects should employ some systematic process to get to your solution and a clearly identified way to evaluate it to ensure it solves the problem statement identified.

The project will follow an iterative and research-driven development process. I will start by conducting background research on multi-signature authentication cryptography primitives, existing authentication systems, and any other relevant security design patterns. That research will then inform application specifications for the proxy and approval flow. I will then build a minimum viable product that implements the core flow of intercepting a request, collecting approvals until a quorum is reached, and forwarding the request to the target endpoint. After reviewing the MVP, I will identify any missing requirements, limitations, and security concerns, then return to do additional research and refinement of the specification before implementing the full system. I will then develop an evaluation system that ensures my project solves the intended use cases I've identified. This evaluation system will likely consist of integration tests that verify that the project works for it's intended use cases (usability), blocks attacks that attempt to bypass authentication mechanisms (security), and quantifies proxy overhead (performance), though optimizations are out of scope for this proof-of-concept.

Timeline

Enter one task per row. Include only project tasks (not peer feedback or progress report submissions). Tasks must be actionable. Update the Status column each week (Completed / In Progress / Cancelled / etc.).

Week	Description of Task	Status
W1	Brainstorm ideas for the project	Complete
W1	Create Project Proposal	Complete
W2	Conduct broad project research	Complete
W2	Refine Problem/Solution Statement and Methodology	Complete
W2	Define Project Timeline	Complete
W2	Conduct research on multi-auth crypto primitives	In Progress
W3	Develop authentication scheme	Not Started
W3	Define high-level architecture and data flow	Not Started
W4	Write detailed application specifications	Not Started
W4	Create harness for AI assisted programming	Not Started
W5	Develop minimum viable product	Not Started
W6	Review MVP to identify missing requirements, limitations, and security concerns	Not Started
W6	Refine application specifications based on MVP review	Not Started
W7	Develop MVP evaluation framework based on use cases, security concerns, and performance	Not Started
W8	Begin building fully featured application based on project specifications	Not Started
W8	Create report outline	Not Started
W9	Continue iterating on application development	Not Started
W9	Begin writing report	Not Started
W10	Evaluate project based on security, usability, and performance	Not Started
W10	Continue writing report	Not Started
W11	Add finishing touches to project implementation	Not Started
W11	Finalize report	Not Started

Evaluation

Include any evaluation plans and/or results by Progress Report 3. This may expand as you finalize the report.

Report Outline

Include an outline of your final report by Progress Report 4. This may expand as you finalize the report.

References

List sources you have reviewed for your project.

Appendix

If there are notes, sources, or figures you want to reference, include them here. Draft work product should begin by Progress Report 3.